

Task-1

Suppose you have this email address “Amit_ml@gmail.edu”

- Input Validation: Check if the input string contains exactly one "@" symbol and at least one "." after the "@" symbol. If it's not a valid email, return "Invalid email".
- Extract Username: Extract and return the part of the email before the "@" symbol.
- Extract Domain: Extract and return the domain (the part between "@" and the last ".").
- Check for Domain Ending: Check if the email ends with ".com". If it does, return "Commercial Domain". If it ends with ".edu", return "Educational Domain". Otherwise, return "Other Domain".

Task-2:

Encoded Message:

[There is a fault in this question](#)

###!!@moclEW EPGTQ!!!6789

Steps to Decode:

1. Extract the core part of the message: "moclEW EPGTQ".
2. Reverse the first word: "moclEW" becomes "Welcome".
3. Replace shifted vowels in the second word:
 - o "EPGTQ": No vowels to change.
4. Final decoded message: "Welcome PGTQ".

Task-3:

Encoded Message:

&&&*\$gnirtS PLIO!!@1234

Steps to Decode:

1. Extract the core part of the message: "gnirtS PLIO".
2. Reverse the first word: "gnirtS" becomes "String".
3. Replace shifted vowels in the second word:
 - o "PLIO": Replace I->E and O->U to get "PLEU".
4. Final decoded message: "String PLEU".

Task-4:

Encoded Message:

##\$\$\$@!yalpstcejorp EPUVT****9887

Steps to Decode:

1. Extract the core part of the message: "yalpstcejorp EPUVT".
2. Reverse the first word: "yalpstcejorp" becomes "projectplay".
3. Replace shifted vowels in the second word:
 - o "EPUVT": Replace E->A, U->O to get "APOVT".
4. Final decoded message: "projectplay APTOV".